

Disproof of Conjecture 00000008579: the eigenvalue multiplicities of the Ornstein–Uhlenbeck operator are not the partition numbers

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Abstract

Conjecture 00000008579 makes four claims about the Ornstein–Uhlenbeck operator on Wiener space: its spectrum is $\{0, 1, 2, \dots\}$; the multiplicity of the eigenvalue n is the partition number $p(n)$; the spectral counting function has Hardy–Ramanujan asymptotics; and its second-order correction is the Rademacher series. We refute the second claim (the “partition multiplicity law”), and hence the conjunction. The eigenvalue n of $N = -L$ has the n -th Wiener chaos as its eigenspace. On a Wiener space with infinitely many coordinates this space is infinite-dimensional for every $n \geq 1$: for example $W(e_0), W(e_1), \dots$ are linearly independent eigenvectors for the eigenvalue 1, while $p(1) = 1$. On the finite-dimensional Gaussian space $\mathbb{R}^d$ the multiplicity is $\binom{n+d-1}{d-1}$. For $d \geq 2$ this gives $\text{mult}(1) = d \geq 2 \neq p(1)$. For $d = 1$ every multiplicity is 1, which differs from $p(n)$ for every $n \geq 2$. So the law fails on every Gaussian space, at $n = 1$ or $n = 2$. The decisive objects are formalized in Lean 4 (core library only): polynomials, ∂_i , multiplication by x_i , the generator $L = \sum_i (\partial_i^2 - x_i \partial_i)$, eigenspaces, linear independence, multiplicity and $p(n)$. The final theorem refutes the law in every dimension $d \geq 0$ and on the infinite-dimensional (cylinder) Wiener space.

1 The conjecture and its precise reading

Definition: The Malliavin spectrum of the Ornstein–Uhlenbeck operator on Wiener space. *Conjecture:* The spectrum consists of the nonnegative integers (a Hermite spectrum law); the eigenvalue multiplicities are the partition numbers of the corresponding degree (a partition multiplicity law); the asymptotics of spectral counting is given by the Hardy–Ramanujan partition asymptotics (an H-R asymptotic law); and the second-order correction is the Rademacher series (an R-correction law).

The Chinese version says the same, clause by clause (“spectrum is the nonnegative integers; and the eigen-multiplicities are the partition numbers of the corresponding degree; ...”). The conjecture is a conjunction of four clauses (1)–(4). We refute

(2) *for every $n \geq 0$, the eigenvalue n of $N = -L$ has multiplicity exactly $p(n)$,*

where $p(n)$ is the number of partitions of n , so $p(0), p(1), p(2), \dots = 1, 1, 2, 3, 5, 7, \dots$. “The corresponding degree” of the eigenvalue n is n , because the eigenspace of n is the n -th Wiener chaos, which consists of Hermite polynomials of degree n . Shifted indexings are treated in Section 4. Refuting one clause of a conjunction refutes the conjunction, so clauses (1), (3) and (4) need not be interpreted. Clause (1) is in fact true.

The objects

Let (Ω, H, μ) be a Gaussian space with isonormal process W on a real Hilbert space H . Examples are classical Wiener space ($H = L^2[0, 1]$, so H is infinite-dimensional) and $\mathbb{R}^d$ with the standard Gaussian measure γ_d ($H = \mathbb{R}^d$, $W(e_i) = x_i$). The Ornstein–Uhlenbeck operator is $L = -\delta D$, where D is the Malliavin derivative and δ its adjoint. Let (e_i) be an orthonormal family in H . On a cylinder polynomial $F = f(W(e_0), \dots, W(e_{d-1}))$ with f a polynomial, L acts by

$$LF = (L_d f)(W(e_0), \dots, W(e_{d-1})), \quad L_d = \sum_{i < d} \left(\frac{\partial^2}{\partial x_i^2} - x_i \frac{\partial}{\partial x_i} \right) = \Delta - x \cdot \nabla. \quad (1)$$

(This is standard; see Nualart, *The Malliavin Calculus and Related Topics*, §1.4.) The map $f \mapsto f(W(e_0), \dots, W(e_{d-1}))$ is injective on polynomials, and it is an isometry from $L^2(\gamma_d)$ into $L^2(\mu)$ because the variables $W(e_i)$ are i.i.d. standard normal. Polynomials lie in the domain of the self-adjoint operator $N = -L$, and N extends $-L_d$ on them. Hence:

Lemma 1. *If $f_1, \dots, f_m$ are linearly independent polynomials with $L_d f_j = -n f_j$, then $F_j = f_j(W(e))$ are m linearly independent eigenvectors of N for the eigenvalue n . In particular $\text{mult}(n) \geq m$, where $\text{mult}(n) = \dim \ker(N - n)$.*

2 Polynomial computations

We identify a polynomial with its coefficient map $\beta \mapsto f_\beta$, where $\beta \in \mathbb{N}^{(\mathbb{N})}$ is a multi-index and $x^\beta = \prod_i x_i^{\beta_i}$. Then $(\partial_i f)_\beta = (\beta_i + 1) f_{\beta + e_i}$ and $(x_i f)_\beta = f_{\beta - e_i}$ (which is 0 if $\beta_i = 0$). So $(x_i \partial_i f)_\beta = \beta_i f_\beta$ and

$$(L_d f)_\beta = \sum_{i < d} \left((\beta_i + 1)(\beta_i + 2) f_{\beta + 2e_i} - \beta_i f_\beta \right). \quad (2)$$

These are exactly the definitions used in Lean (`deriv`, `mulX`, `L`, `L.apply`). As a sanity check, Lean also proves the canonical commutation relation $[\partial_i, x_i] = 1$.

Lemma 2 (squarefree monomials). *If $\alpha_i \leq 1$ for all i and $\alpha_i = 0$ for $i \geq d$, then $L_d x^\alpha = -|\alpha| x^\alpha$ with $|\alpha| = \sum_i \alpha_i$.*

Proof. $\partial_i^2 x^\alpha = 0$ since $\alpha_i \leq 1$, and $x_i \partial_i x^\alpha = \alpha_i x^\alpha$. Sum over $i < d$. □

In particular $L_d x_i = -x_i$ for $i < d$, and $L_d(x_0 x_1 \cdots x_{n-1}) = -n x_0 \cdots x_{n-1}$. Distinct monomials are linearly independent.

Theorem 3 ($d \geq 2$). *On $\mathbb{R}^d$ with $d \geq 2$, $\text{mult}(1) \geq d \geq 2 > 1 = p(1)$. More generally $\text{mult}(n) \geq \lfloor d/n \rfloor$ for $n \geq 1$.*

Proof. The monomials $x_0, \dots, x_{d-1}$ are independent eigenvectors for the eigenvalue 1 (Lemma 2, Lemma 1). For general n , the block monomials $b_j = x_{nj} x_{nj+1} \cdots x_{nj+n-1}$ with $0 \leq j < \lfloor d/n \rfloor$ are distinct squarefree monomials of degree n . □

Theorem 4 (Wiener space). *If $\dim H = \infty$ (for example on classical Wiener space), then every eigenvalue $n \geq 1$ of N has infinite multiplicity, so $\text{mult}(n) \neq p(n)$ for every $n \geq 1$.*

Proof. Choose an orthonormal sequence $e_0, e_1, \dots$ in H . For every m , the functionals $\prod_{k < n} W(e_{nj+k})$ with $j < m$ are m independent eigenvectors (Theorem 3 with $d = nm$, and Lemma 1). For $n = 1$ these are just $W(e_0), \dots, W(e_{m-1})$. □

Theorem 5 ($d = 1$). *On $(\mathbb{R}, \gamma_1)$ every eigenvalue n has multiplicity exactly 1, with eigenspace $\mathbb{R}H_n$ (H_n the probabilists' Hermite polynomial). Hence $\text{mult}(n) = 1 < 2 \leq p(n)$ for every $n \geq 2$. In particular, for $n = 2$ the eigenvectors are exactly the multiples of $x^2 - 1$, while $p(2) = 2$.*

Proof. Polynomial eigenvectors. Write $g_k = f_{(k)}$ for the coefficient of x^k . By (2), $L_1 f = -nf$ is equivalent to

$$(k+1)(k+2)g_{k+2} = (k-n)g_k \quad (k \geq 0). \quad (3)$$

Suppose g is finitely supported, satisfies (3), and $g_n = 0$. For $k \neq n$, (3) shows that $g_{k+2} = 0$ implies $g_k = 0$. Descending from the top of the support gives $g_k = 0$ for all $k > n$. Then $g_{n+1} = 0$ and $g_n = 0$, and descending from there gives $g_k = 0$ for $k < n$. So $g = 0$. If f and f' are two eigenvectors, then $f'_n f - f_n f'$ is an eigenvector with vanishing x^n -coefficient, hence zero. So any two eigenvectors are proportional, and the polynomial eigenspace has dimension ≤ 1 . It contains H_n , which satisfies $H''_n - xH'_n = -nH_n$; for example $L_1(x^2 - 1) = 2 - 2x^2$. *All L^2 eigenvectors.* The Hermite polynomials H_k form an orthogonal basis of $L^2(\gamma_1)$ and $NH_k = kH_k$. If $NF = nF$, then for each k , $k\langle F, H_k \rangle = \langle F, NH_k \rangle = \langle NF, H_k \rangle = n\langle F, H_k \rangle$ by self-adjointness. So $\langle F, H_k \rangle = 0$ for $k \neq n$, and $F \in \mathbb{R}H_n$. Finally, for $n \geq 2$ the partitions (n) and $(1, \dots, 1)$ are distinct, so $p(n) \geq 2$. $\square$

Theorem 6 (exact multiplicities). *On $(\mathbb{R}^d, \gamma_d)$, $\text{mult}(n) = \binom{n+d-1}{d-1}$, the number of multi-indices $\alpha \in \mathbb{N}^d$ with $|\alpha| = n$.*

Proof. The products $H_\alpha(x) = \prod_i H_{\alpha_i}(x_i)$ form an orthogonal basis of $L^2(\gamma_d)$, and $NH_\alpha = |\alpha|H_\alpha$ because L_d is a sum of one-variable operators. The argument of Theorem 5 then shows that the eigenspace of n is $\text{span}\{H_\alpha : |\alpha| = n\}$. $\square$

The table below is recomputed by `verify.py` as kernel dimensions of $L_d + n$ on polynomials of degree $\leq n$ (and, as a check, $\leq n+2$):

n	0	1	2	3	4	5
$p(n)$	1	1	2	3	5	7
$d = 1$	1	1	1	1	1	1
$d = 2$	1	2	3	4	5	6
$d = 3$	1	3	6	10	15	21
$d = 4$	1	4	10	20	35	56

Note the isolated coincidence $d = 2, n = 4$ ($5 = p(4)$). The law already fails there at $n = 1, 2, 3$.

3 Main theorem

Theorem 7. *Clause (2) is false on every Gaussian space: on $\mathbb{R}^d$ for every $d \geq 0$, and on every Wiener space with $\dim H = \infty$. Hence Conjecture 00000008579 is false.*

Proof. For $d = 0$ (constants only) there is no eigenvector for $n = 1$, but $p(1) = 1$. For $d = 1$ use Theorem 5 at $n = 2$, for $d \geq 2$ Theorem 3 at $n = 1$, and for $\dim H = \infty$ Theorem 4 at $n = 1$. In every case the second clause fails, so the conjunction fails. $\square$

4 Robustness: other readings

1. *Which Wiener space.* Classical Wiener space, any abstract Wiener space, or any Gaussian Hilbert space of infinite dimension: Theorem 4. Finite-dimensional Gaussian spaces: Theorems 3 and 5. The answer does not depend on the choice of H , apart from its dimension.
2. *Field and kind of multiplicity.* N is self-adjoint, so algebraic and geometric multiplicities agree. Complexifying does not change dimensions. Lean measures rank over $\mathbb{Z}$, which equals the dimension over $\mathbb{Q}$ (clear denominators) and over $\mathbb{R}$ for the rational polynomials used.
3. *Shifted index* ($\text{mult}(n) = p(n + s)$). For $d = 1$, $1 \neq p(n + s)$ as soon as $n + s \geq 2$. For $d \geq 2$ the reading fails by item 4, and `verify.py` finds an explicit failure for $s \in \{-1, 1, 2\}$ and all $d \leq 29$. On infinite-dimensional Wiener space the multiplicities are infinite.
4. *“For all large n ” / asymptotic readings.* For finite d , $\binom{n+d-1}{d-1}$ is a polynomial of degree $d - 1$ in n . Meanwhile $p(n) \geq \binom{\lfloor n/(d+1) \rfloor + d}{d}$: choose multiplicities $m_2, \dots, m_{d+1} \geq 0$ with $\sum m_i \leq \lfloor n/(d+1) \rfloor$, so that $\sum_i i m_i \leq n$, and fill up with parts 1. This lower bound is a polynomial of degree d , so $p(n) > \text{mult}(n)$ for all large n ; e.g. for $d = 1, 2, 3$, for all $n > 1, 4, 13$. The same comparison refutes clause (3) for finite d : the counting function $\#\{\text{eigenvalues} \leq \lambda\} = \binom{\lfloor \lambda \rfloor + d}{d}$ is a polynomial, not of Hardy–Ramanujan type. For $\dim H = \infty$ every multiplicity is infinite.
5. *Restricting to symmetric functionals.* If one keeps only eigenvectors invariant under permutations of the d coordinates, the multiplicity of n is the number of partitions of n into at most d parts. This equals $p(n)$ only for $n \leq d$, so for each fixed d it fails at $n = d + 1$ (checked in `verify.py` for $d \leq 3$). With infinitely many i.i.d. coordinates, the Hewitt–Savage 0–1 law makes every exchangeable L^2 functional a.s. constant, so the multiplicities are 0 for $n \geq 1$.
6. *The weighted reading: is there any reading that gives $p(n)$?* Yes, for a different operator. Take the second quantization $d\Gamma(A)$ of $A = \text{diag}(1, 2, 3, \dots)$, i.e. $\sum_k k(\partial_k^2 - x_k \partial_k)$ in infinitely many coordinates. This is an Ornstein–Uhlenbeck operator with drift $-A$, or a string-oscillator energy operator. It has eigenvalue $\sum_k k \alpha_k$ on H_α , so its multiplicities are exactly $p(n)$. `verify.py` checks this for $d = 4, 5$ and $n \leq d$. **For $d\Gamma(A)$, clause (2) holds**, so our disproof is specific to the operator the statement names.

The strongest argument for this reading is the following. Clauses (3) and (4), Hardy–Ramanujan counting and the Rademacher correction, only make sense if the multiplicities are finite. For $N = d\Gamma(I)$ on Wiener space they are infinite (Theorem 4), so a reader might conclude that the author must have meant $d\Gamma(A)$.

We still do not adopt this reading, for two reasons.

- *Definition.* In Malliavin calculus “the Ornstein–Uhlenbeck operator” is by definition $L = -\delta D$, i.e. $N = -L = d\Gamma(I)$, the number operator (Nualart, §1.4). Its eigenspaces are the Wiener chaoses, and the “Hermite spectrum law” of clause (1) refers to exactly these Hermite eigenfunctions.
- *Canonicity.* On classical Wiener space there is no canonical operator A with spectrum $\{1, 2, 3, \dots\}$ to quantize. The natural candidates are the Brownian covariance operator (kernel $\min(s, t)$ on $L^2[0, 1]$) and the Cameron–Martin Laplacian ($-d^2/dt^2$ with $h(0) = 0$, $h'(1) = 0$). Their eigenvalues are $((k - \frac{1}{2})\pi)^{-2}$ and $((k - \frac{1}{2})\pi)^2$ respectively, and neither is an integer. Choosing $A = \text{diag}(1, 2, 3, \dots)$ requires an extra, non-canonical choice of basis and weights that the statement does not make.

We therefore regard the weighted operator as a different conjecture, not a reading of this one. Under the named operator, clauses (3) and (4) also fail or are meaningless (item 4 above).

5 Lean formalization

The Lean 4.19.0 project in `lean4/` uses the core library only. Its definitions are as follows.

- `Poly := (Nat -> Nat) -> Int`, the coefficient map, and `mono, x i, H2 =  $x_0^2 - 1$` .
- The predicates `InVars`, `Bounded` and `IsPoly d` (finite support in $x_0, \dots, x_{d-1}$).
- `deriv, mulX`, and `L d f = fun b => isum d (fun i => deriv i (deriv i f) b - mulX i (deriv i f) b)`.
- `Eig d n f` (f is a polynomial with $Lf = -nf$), `LinIndep`, `MultGE d n m`, `Mult d n m` ($\geq m$ and not $\geq m + 1$), and the cylinder versions `EigCyl`, `MultGECyl`, `MultCyl`.
- `IsPartition`, `partitionsL, p`, with `mem_partitionsL` for all n .

The main theorems are as follows.

- `L.apply` (formula (2)) and `deriv_mulX_comm` ($[\partial_i, x_i] = 1$).
- `Lx0_x1` ($Lx_0 = -x_0$, $Lx_1 = -x_1$) and `linIndep_x`.
- `eig_H2` ($L(x^2 - 1) = -2(x^2 - 1)$) and `multGE.blocks` ($\text{mult}_d(n) \geq m$ when $nm \leq d$).
- `eig1_prop`: in one variable, any two eigenvectors for the same eigenvalue, of any degree, are proportional. Consequences: `mult1_le` ($\text{mult}_1(n) \leq 1$ for all n), `eig2_eq_H2` (every eigenvector for 2 is $f_2 \cdot (x^2 - 1)$), and `mult1_exact` ($\text{mult}_1(0) = \text{mult}_1(1) = \text{mult}_1(2) = 1$).
- `p_values` ($p(0..5) = 1, 1, 2, 3, 5, 7$) and `p_ge_two`.
- `partitionLaw_false d`: $\neg \text{PartitionLaw } d$ for every d ; `d1_fails_all` (for $d = 1$ the law fails at every $n \geq 2$); `mult_one_ge_two`.
- `L_eq_of_isPoly`: L is consistent across dimensions, so it is well defined on cylinder polynomials. `multCyl_infinite` (no finite multiplicity for any $n \geq 1$) and `partitionLawCyl_false`.
- `conjecture_00000008579_false` and `conjecture_00000008579_false_wiener`: $\neg(C_1 \wedge (2) \wedge C_3 \wedge C_4)$ for arbitrary propositions C_1, C_3, C_4 .

The project contains no `sorry`, no `native_decide` and no added axioms. `#print axioms` reports `propext`, `Classical.choice` and `Quot.sound`. Classical choice is used only to decide equality of multi-indices, which are functions, in the definition of monomials.

Limitations. Lean works on the polynomial core and does not formalize $L^2(\mu)$, the closure of L , or Hermite completeness. For $d \geq 2$ and for Wiener space the disproof needs only lower bounds, and Lemma 1 transfers polynomial eigenvectors to genuine eigenvectors. For $d = 1$ it needs the upper bound $\text{mult}(2) \leq 1$. Lean proves this bound for the polynomial eigenspace, and the extension to all of $L^2(\gamma_1)$ (Hermite completeness) is Theorem 5, proved here only. Existence of H_n for general n (so that $\text{mult}_1(n) = 1$ exactly) and the exact formula of Theorem 6 are proved in this report and checked numerically in `verify.py`. Lean proves $\text{mult}_1(n) = 1$ exactly only for $n \leq 2$, and for $d \geq 2$ it proves lower bounds only. The readings in Section 4, items 3–6, are covered by this report and `verify.py`, except for $d = 1$ at every $n \geq 2$, which is in Lean.

6 Independent check

`verify.py` (Python 3 standard library) uses a different method. It builds the matrix of L_d on the monomial basis and computes kernel dimensions of $L_d + n$ exactly over $\mathbb{Q}$. It checks the following.

- The table above for $d \leq 4$, $n \leq 5$, and that degree $\leq n + 2$ gives the same dimensions.
- $\text{mult}_d(1) = d$ for $d \leq 7$.
- L of the Hermite polynomials for $n \leq 12$.
- $p(n)$ by Euler's recurrence and by enumeration.
- The shifted, eventual, symmetric and weighted-operator readings of Section 4.

All checks pass.